

Quiz 2

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz.

Problem 1. (6 points) Given two vectors $\mathbf{u} = \langle 4, 0, 3 \rangle$ and $\mathbf{v} = \langle 3, -2, 6 \rangle$.

(a) Find the vector projection of $\mathbf{u}$ onto $\mathbf{v}$, $\text{proj}_{\mathbf{v}} \mathbf{u}$.

$$\mathbf{u} \cdot \mathbf{v} = \langle 4, 0, 3 \rangle \cdot \langle 3, -2, 6 \rangle = 4 \cdot 3 + 0 \cdot (-2) + 3 \cdot 6 = 30$$

$$|\mathbf{v}| = \sqrt{3^2 + (-2)^2 + 6^2} = \sqrt{49} = 7$$

$$\text{Proj}_{\mathbf{v}} \mathbf{u} = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2} \right) \mathbf{v} = \left(\frac{30}{49} \right) \langle 3, -2, 6 \rangle$$

$$= \left\langle \frac{90}{49}, -\frac{60}{49}, \frac{180}{49} \right\rangle$$

(b) Compute $\mathbf{u} \times \mathbf{v}$ and then find the area of the parallelogram determined by $\mathbf{u}$ and $\mathbf{v}$.

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & 0 & 3 \\ 3 & -2 & 6 \end{vmatrix} = \mathbf{i} \begin{vmatrix} 0 & 3 \\ -2 & 6 \end{vmatrix} - \mathbf{j} \begin{vmatrix} 4 & 3 \\ 3 & 6 \end{vmatrix} + \mathbf{k} \begin{vmatrix} 4 & 0 \\ 3 & -2 \end{vmatrix}$$

$$= [0 \cdot 6 - 3 \cdot (-2)] \mathbf{i} - [4 \cdot 6 - 3 \cdot 3] \mathbf{j} + [4 \cdot (-2) - 0 \cdot 3] \mathbf{k}$$

$$= 6\mathbf{i} - 15\mathbf{j} - 8\mathbf{k}$$

$$\text{Area of parallelogram} = |\mathbf{u} \times \mathbf{v}| = \sqrt{6^2 + (-15)^2 + (-8)^2} = \sqrt{325} = 13\sqrt{5}$$

Problem 2. (4 points)

(a) Find parametric equations for the line through the points $P(1, 2, 3)$ and $Q(6, 3, 5)$.

$$\text{Line direction: } \mathbf{v} = \overrightarrow{PQ} = \langle 5, 1, 2 \rangle$$

Parametric equations

$$x = 1 + 5t, \quad y = 2 + t, \quad z = 3 + 2t$$

$$-\infty < t < \infty$$

(b) Find an equation for the plane through $P(0, 2, -1)$ normal to the vector $3\mathbf{i} - 2\mathbf{j} - \mathbf{k}$.

$$3(x - 0) - 2(y - 2) - (z - (-1)) = 0$$

$$3x - 2y + 4 - z - 1 = 0$$

$$3x - 2y - z = -3$$